Euclid's Elements — Book I
Proposition 23

Formal Reconstruction — Technical Documentation

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CGJteamLab

On a given straight line, and at a given point on it, to construct a rectilinear angle congruent to a given rectilinear angle.

1 Introduction

Euclid's Proposition I.23 asks for the construction, at a prescribed point on a prescribed straight line, of an angle congruent to a given rectilinear angle.

In Euclid's development this is a genuine construction theorem. The construction proceeds indirectly: Euclid chooses points on the arms of the given angle, obtains three segment lengths, uses Proposition I.22 to construct a triangle with those three sides, and then invokes Proposition I.8 (SSS) to identify the required angle.

In the Hilbert reconstruction used by CGJteamLab, the logical status of the proposition is different. Hilbert's congruence axiom III.4 already provides angle transport on a prescribed side of a prescribed ray, together with uniqueness of the resulting ray. Consequently the formal Proposition I.23 is a short wrapper over a stronger foundational principle.

This contrast is mathematically important. Proposition I.23 is an example in which the order of Euclid's propositions and the dependency order induced by the chosen axiomatic foundation no longer coincide.

2 The Classical Configuration

Euclid's construction is best understood in the original notation of the classical argument. A straight segment AB and a given angle at C are prescribed. The objective is to construct at A, on the given straight line, an angle equal to the angle at C.

Wilkins's presentation makes the constructional character of the proposition especially clear. The initial data are simply the line segment and the given angle:

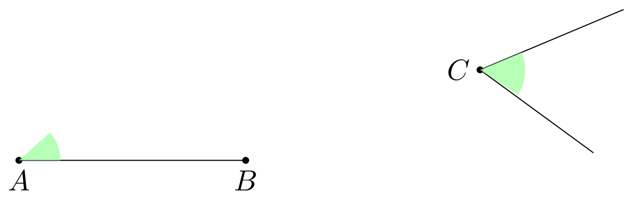

Figure 1: Initial data for Euclid I.23: a prescribed segment AB and a given rectilinear angle at C.

Euclid does not transport the angle directly. He first chooses points D and E on the two arms of the given angle and forms the triangle CDE. A second triangle AFG is then constructed with G lying on AB.

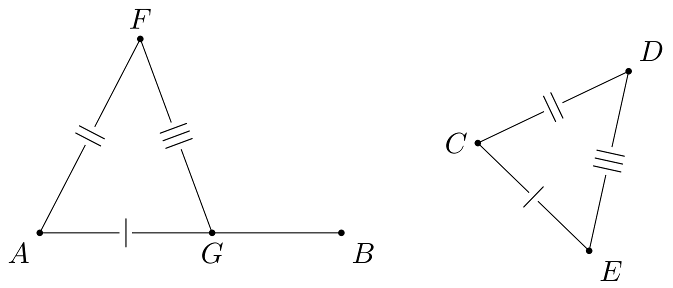

Figure 2: Euclid's completed auxiliary construction: $AG \cong CE$, $AF \cong CD$, and $FG \cong DE$.

The three side congruences give

$$\triangle AFG \cong \triangle CDE$$

by Proposition I.8, and hence

$$\angle FAG \cong \angle DCE.$$

The drawings therefore encode the actual Euclidean construction rather than merely displaying two congruent angles.

3 Classical Proof

Euclid derives the construction from earlier propositions rather than treating angle transport as primitive.

Choose points A and C on the two arms of the given angle $\angle ABC$. The three segments

$$AB, \qquad BC, \qquad AC$$

determine the triangle ABC.

Now use Proposition I.22 to construct a triangle DEF whose corresponding sides satisfy

$$DE \cong AB, \qquad DF \cong BC, \qquad EF \cong AC.$$

By Proposition I.8, the two triangles are congruent by SSS. Hence their corresponding included angles are congruent, and in particular

$$\angle ABC \cong \angle EDF$$

after the appropriate correspondence of vertices.

Thus the classical dependency pattern is

$$\begin{gathered}\text{given angle}\\ \downarrow\\ \text{choose a triangle containing the angle}\\ \downarrow\\ \text{I.22}\\ \downarrow\\ \text{construct a triangle with the same three side lengths}\\ \downarrow\\ \text{I.8 (SSS)}\\ \downarrow\\ \text{corresponding angle congruence}\\ \downarrow\\ \text{I.23.}\end{gathered}$$

In Euclid's route, angle transport is therefore reduced to segment construction and triangle congruence.

4 Hilbert III.4

Hilbert reverses the foundational emphasis.

Instead of deriving angle transport from triangle construction, Hilbert includes angle construction among the congruence axioms. In the present formal layer the relevant operation is

```
HilbertCongruence.angle_construction
```

Conceptually, it states that from

- a nondegenerate source angle;
- a prescribed initial ray;
- a choice of one side of the supporting line,

one obtains a ray on that side forming an angle congruent to the source angle.

The Hilbert axiom is stronger than Euclid I.23 because it also contains a uniqueness assertion: on the chosen side of the line, the constructed ray is unique.

This additional information is not needed in the public statement of Proposition I.23, but it remains available in the Hilbert interface.

5 Proof Architecture of the Reconstruction

The formal reconstruction separates naturally into four stages.

5.1 The Source Angle

The hypothesis

$$\neg\,\mathrm{Collinear}(A, B, C)$$

ensures that $\angle ABC$ is nondegenerate.

This is the source angle that will be transported.

5.2 The Prescribed Ray

The points D and E determine the initial ray of the new angle.

The hypotheses assert that

$$D \neq E$$

and that both points lie on a common supporting line. Thus the formal data specify not merely a vertex, but an oriented ray represented by two distinct points.

5.3 Selecting the Side

A point S not lying on the supporting line selects one of its two sides.

The constructed point F is required to satisfy

$$\mathrm{SameSide}(F, S, \mathrm{base}).$$

This datum has no explicit counterpart in Euclid's informal formulation, where the intended side is supplied by the diagram and by the instruction to construct the angle "on" a given straight line.

In the axiomatic reconstruction it must be stated.

5.4 Applying Angle Construction

The Hilbert angle-construction theorem supplies a point F satisfying the required same-side condition and

$$\angle ABC \cong \angle DEF.$$

Its uniqueness component is discarded because Proposition I.23 asks only for existence.

Thus the entire formal proof is an interface extraction from Hilbert III.4.

The Hilbert configuration is consequently much simpler than the Euclidean construction. The source angle, the prescribed ray DE, and the side selector S determine the required ray directly:

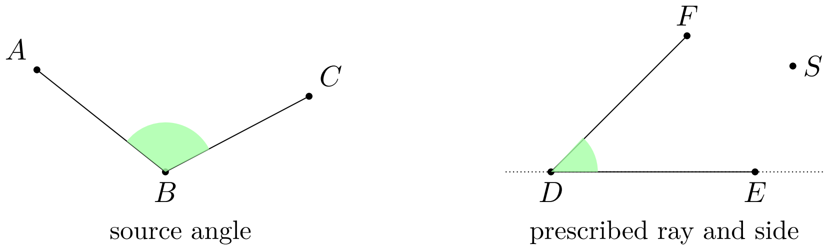

Figure 3: Hilbert III.4 configuration. The point S selects the side of the supporting line on which the second ray is constructed.

This diagram is intentionally not the Euclidean CDE/AFG construction. It represents the stronger primitive available in the Hilbert congruence layer.

6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_23
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E S : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hDE : D != E)
    (base : Geo.Line)
    (hDbase : HilbertIncidence.OnLine D base)
    (hEbase : HilbertIncidence.OnLine E base)
    (hSbase : Not (HilbertIncidence.OnLine S base)) :
    Exists fun F : Geo.Point =>
      HilbertSameSide Geo F S base /\
      Geo.AngleCongruent A B C D E F := by
```

The mathematical meaning of the arguments is:

$$\angle ABC$$

is the source angle,

$$DE$$

is the prescribed initial ray, and S selects the side of the supporting line on which the second ray must be constructed.

The proof then invokes the Hilbert construction principle and extracts the components required by the Euclidean statement. Schematically:

```
rcases HilbertCongruence.angle_construction ... with
  <F, hSameSide, hAngle, hUnique>

exact <F, hSameSide, hAngle>
```

The uniqueness witness `hUnique` is deliberately not returned.

This is the essential feature of the Lean proof: its brevity reflects the strength of the Hilbert interface rather than an omitted geometric argument.

7 The Euclidean Route and the Hilbert Route

7.1 The Euclidean Route

Euclid reduces angle transport to triangle construction:

$$\mathrm{I.22} + \mathrm{I.8} \longrightarrow \mathrm{I.23}.$$

Proposition I.22 constructs a triangle with prescribed side lengths, while Proposition I.8 identifies the corresponding angle by SSS.

The contrast is especially transparent in dependency form:

```
Euclid
------
given angle at C
     |
choose D,E on its arms
     |
triangle CDE
     |
I.20  -> triangle inequalities
     |
I.22  -> construct AFG
     |
I.8 / SSS
     |
angle FAG ~= angle DCE


Hilbert / Lean
--------------
source angle + prescribed ray + side selector
     |
Hilbert III.4
     |
constructed ray + angle congruence
     |
euclid_proposition_23
```

This route passes through segment arithmetic, circle intersection, continuity, and triangle congruence.

7.2 The Hilbert Route

In the Hilbert foundation the corresponding dependency is simply

$$\text{III.4} \longrightarrow \text{I.23.}$$

Angle transport has already been placed in the congruence layer.

Thus Proposition I.23 does not formally depend on Proposition I.22 or Proposition I.8.

7.3 Foundational Interpretation

The difference is not merely one of proof length.

Euclid treats angle transport as something to be constructed from more elementary operations. Hilbert treats the ability to copy an angle to a prescribed side of a ray as part of the primitive congruence structure.

The same geometric statement therefore occupies two different logical levels in the two systems.

This is precisely the kind of distinction that becomes visible when Book I is reconstructed over an explicit axiomatic interface.

8 Relationship with Earlier Propositions

8.1 Proposition I.8

In Euclid's proof, Proposition I.8 is the final congruence theorem. Once the auxiliary triangle has been constructed with the required three side lengths, SSS identifies the transported angle.

In the formal Hilbert proof I.8 is not used.

8.2 Proposition I.22

Euclid uses Proposition I.22 to construct the auxiliary triangle with prescribed sides.

The formal Hilbert proof again does not use it. Reintroducing I.22 would reconstruct Euclid's historical dependency chain on top of a foundation that already contains the stronger angle-construction principle.

This is especially noteworthy because I.22 required explicit continuity machinery in the reconstruction, whereas I.23 requires none.

8.3 Book I Order versus Formal Dependency

The file for Proposition I.23 naturally follows Proposition I.22 in the Book I sequence.

The logical dependency graph, however, does not contain an edge

$$\text{I.22} \longrightarrow \text{I.23}.$$

The textual order of the *Elements* and the dependency order induced by Hilbert geometry therefore diverge at this point.

9 Hilbert and Book Zero Components Used

9.1 HilbertCongruence.angle_construction

This is the central ingredient of the proof.

It already contains the existence of the required point, the side condition, the angle congruence, and a uniqueness property stronger than the conclusion of I.23.

9.2 HilbertSameSide

The same-side relation makes explicit which of the two mirror-image constructions is intended.

The point S serves as a side selector, and the constructed point F is required to lie on the same side of the supporting line.

9.3 HilbertSameRay

The uniqueness clause of Hilbert III.4 is naturally expressed in terms of equality of rays. In the project, rays are represented through points together with the relation `HilbertSameRay` rather than as a separate primitive type.

9.4 No New Book Zero Lemma

Proposition I.23 exposes no missing low-level operation.

Unlike several preceding propositions, it does not require a new Book Zero theorem. The needed construction already exists at the Hilbert congruence level.

This negative result is structurally useful: a Book I proposition should not generate helper infrastructure merely because it appears as a separate proposition in Euclid.

10 Uniqueness and the Strength of the Hilbert Statement

Euclid asks for a construction.

Hilbert III.4 gives existence together with uniqueness on the selected side of the supporting line.

Suppose another point F' lies on the selected side and determines an angle congruent to the same source angle. The Hilbert uniqueness clause says that the ray from the new vertex through F' coincides with the ray through F.

Thus the formal public theorem intentionally forgets information.

This is a useful interface-design principle: the theorem representing Euclid I.23 should state exactly the Euclidean conclusion, while the stronger foundational theorem remains available separately.

11 Dependency Summary

The formal dependency structure is

$$\begin{gathered}\text{Hilbert incidence + congruence}\\ \downarrow\\ \text{nondegenerate source angle}\\ \text{prescribed ray and supporting line}\\ \text{side selector}\\ \downarrow\\ \texttt{HilbertCongruence.angle\_construction}\\ \downarrow\\ F \text{ on the prescribed side}\\ \angle ABC \cong \angle DEF\\ \downarrow\\ \texttt{euclid\_proposition\_23}.\end{gathered}$$

No use is made of

$$\text{I.22}, \qquad \text{I.8},$$

or of a new continuity argument.

No new geometric axiom and no new Book Zero lemma are introduced.

12 Formalization Notes

Several features of the formal statement deserve emphasis.

First, the point S is additional explicit data. It selects one side of the line containing D and E. Without this information there are two mirror-image rays satisfying the angle-congruence requirement.

Second, the conditions that D and E lie on the base line and that $D \neq E$ ensure that the prescribed initial ray is nondegenerate.

Third, the hypothesis that A, B, C are noncollinear ensures that the source angle is nondegenerate.

Fourth, the theorem returns a point F, not an abstract ray. This matches the point-based representation of rays used throughout the Hilbert layer.

Finally, the short proof should not be interpreted as evidence that Proposition I.23 is mathematically trivial. Rather, its mathematical content has already been absorbed into the axiomatic foundation.

13 Conclusion

Proposition I.23 provides a particularly clean example of the difference between historical construction order and formal axiomatic dependency.

Euclid constructs the transported angle indirectly. Proposition I.22 produces a triangle with prescribed side lengths, and Proposition I.8 identifies the desired angle by SSS.

Hilbert places the corresponding construction directly in the congruence axioms. The formal reconstruction can therefore obtain Euclid's conclusion immediately from III.4.

The important result is not merely that the Lean proof is short. The reconstruction identifies exactly *why* it is short: a theorem in Euclid has become part of the foundational interface in Hilbert geometry.

This also explains why Proposition I.23 introduces no new Book Zero machinery. The proposition does not reveal a missing elementary operation; instead, it reveals a change in foundational organization.

The completed formal theorem therefore preserves Euclid's visible Book I milestone while recording a substantially different dependency graph underneath it.